

Synthetic Empathy Across Cultures:

MULTILINGUAL DETECTION OF EMPATHY IN HINDI AND TELUGU MENTAL HEALTH DIALOGUES



WHY?

- **Highest absolute number of suicides globally**
(NCRB, 2022)
- **Treatment gap: ~83% of people with mental disorders in India receive no treatment** (WHO/ICMR reports)
- Unique empathy expressions shaped by **Indian family, religion, and social norms**



image: <https://www.tpr.org/show/the-source/2018-08-19/how-can-we-change-minds-about-mental-health>

HOW?

Synthetic Data Generation

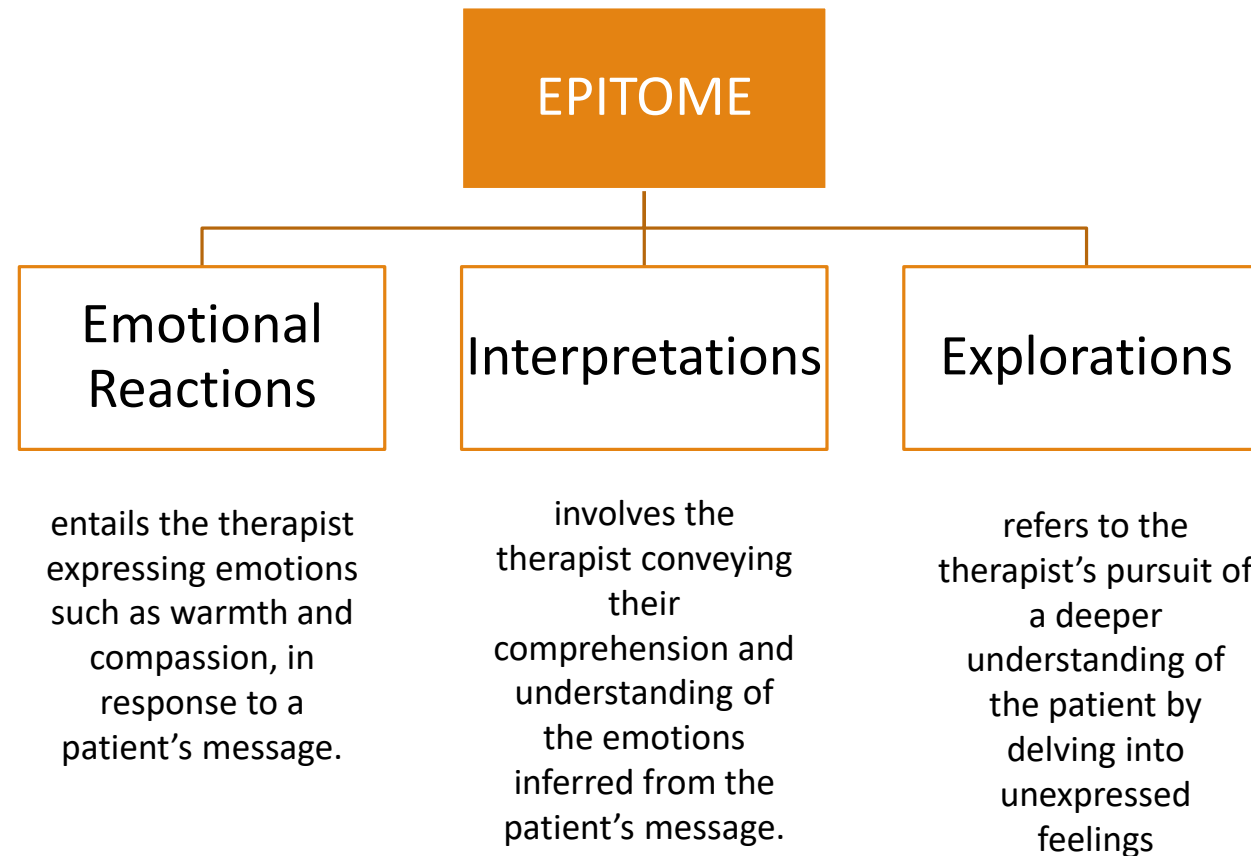
- GPT-3 & Falcon-7B used with empathy prompts
- Reflexion framework adds *verbal feedback* from clinical psychologist (
- Produced **10,464 synthetic therapy dialogues**

Annotation

- 3 trained annotators, guided by EPITOME framework
- Labels: **0 = none, 1 = weak, 2 = strong** empathy
- Agreement (Cohen's κ): Reddit 0.78, GPT-3 synthetic 0.67, Alexander 0.63, MOST+ 0.61

Empathy Classifier

- **RoBERTa-based bi-encoder**, 3 separate models (emotional reaction, interpretation, exploration)
- Output: empathy score (0–2)



Each of these empathy dimensions can take a value of 0, indicating that the therapist is not expressing it at all; 1, indicating a weak degree of expression; or 2, indicating a strong expressing by the therapist. (Sharma et al., 2020)

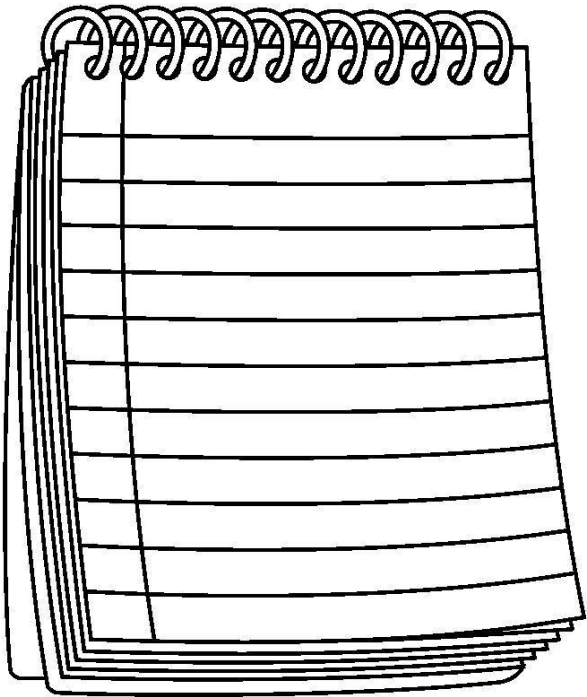
HOW?

Experiments

- **Augmentation:** Add synthetic to Reddit (up to +420 samples)
- **Substitution:** Replace Reddit with 10–50% synthetic
- Evaluation on **579 dialogues** (Alexander + MOST+)
- Metrics: **Accuracy & F1** (per empathy dimension)
- Training: 4 epochs, LR 2e-5, batch size 32, on NVIDIA A100

(Lozoya et al., 2025)

THANK YOU



References

- <https://www.ncrb.gov.in/accidental-deaths-suicides-in-india-year-wise.html>
- <https://pmc.ncbi.nlm.nih.gov/articles/PMC9045352/>
- Ashish Sharma, Adam Miner, David Atkins, and Tim Althoff. 2020. A computational approach to understanding empathy expressed in text-based mental health support. In Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), pages 5263–5276, Online. Association for Computational Linguistics.
- Daniel Lozoya, Michael C. Frank, and Adam Miner. 2025. Synthetic Empathy: Generating Empathic Therapist Responses with Large Language Models. In Proceedings of the 2025 Workshop on Computational Linguistics and Clinical Psychology (CLPsych), pages 109–119, Mexico City, Mexico. Association for Computational Linguistics.